Differential Group Delay Monitoring Used as Feedforward Information for Polarization Mode Dispersion Compensation

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***Abstract*—A novel differential-group-delay (DGD) monitoring scheme that can provide feedforward information to facilitate the compensation of polarization mode dispersion (PMD) is demonstrated. Using a polarization scrambler at the beginning of the link, the instantaneous DGD can be monitored at the end of the link by using a second polarization scrambler followed by a fixed 0.8-dB polarization dependent loss (PDL) element or a polarizer. As the PMD value along the link increases, the instantaneous degree-of-polarization will decrease, thereby resulting in a decrease in the measured RMS power fluctuation value through the PDL element. The authors show instantaneous DGD monitoring of up to ~70 ps for both nonreturn-to-zero (NRZ) and return-to-zero data formats, and they demonstrate PMD compensation of a 10-Gb/s NRZ channel using the monitor output as part of the control feedforward signal.**

***Index Terms*—Differential group delay, optical communications, optical polarization, polarization mode dispersion.**

I. Introduction

One of the critical challenges for next-generation high bit-rate optical fiber transmission systems ($\geq$10 Gb/s/channel) is polarization mode dispersion (PMD) [1]. As the birefringence of a fiber changes randomly along a fiber link and the state-of-polarization (SOP) of an optical signal changes with environmental conditions, PMD effects on the data signal are stochastic and time varying. Therefore, any PMD compensator at a receiver must dynamically track the degrading effects due to PMD and compensate for it. Typical first-order PMD compensators consist of a polarization controller followed by a single fixed or variable differential-group-delay (DGD) element [2], [3]. For the latter case, both the polarization control and the DGD value of the compensator require accurate monitoring information.

Major previously reported PMD monitoring techniques include either measuring the radio frequency (RF) power or analyzing the frequency spectrum of the detected signal [2], [4], or measuring the signal's degree-of-polarization (DOP) [3], [5], [6]. A major problem for the above monitoring schemes is that the effective PMD value depends on both the DGD value and the principal-states-of-polarization (PSP), thereby requiring the feedback incorporating a complicated algorithm to control the polarization controller and variable DGD on a millisecond time scale [7]. Moreover, the link DGD can be ambiguous under the circumstance in which the transmitter SOP is aligned with one of the link input PSPs, thereby causing feedback fading during the tracking procedure.

Recently, polarization scrambling of the input signal's SOP has been used to aid in PMD compensation schemes [8]–[10] as well as monitors that can provide either DGD or PSP information [9], [10]. A key advantage of polarization scrambling at the transmitter is that it can decouple the dependence of the monitored PMD on the input signal's SOP, thus reducing the complexity and increasing the stability of the feedback control.

We propose and demonstrate a simple technique for DGD monitoring using: 1) a 50-kHz periodic polarization scrambler at the transmitter and 2) the combination of a second polarization scrambler, an optical filter, and a polarization-dependent-loss (PDL) element at the receiver. The monitor signal of the instantaneous DGD is generated by the root-mean-square (RMS) value of the optical power that is fluctuating due to the polarization scrambling and subsequent transmission through the PDL element. We measure the instantaneous DGD values up to ~70 ps for both NRZ and RZ 10-Gb/s data formats. Furthermore, as opposed with previous feedback monitoring for which the PMD or DGD monitored values incorporate both the link and compensator contributions, we demonstrate a PMD compensation configuration that uses: 1) our DGD monitor as feedforward signal to preset the value of the tunable DGD element and 2) a traditional bit-error-rate (BER) monitor scheme that provides feedback to the polarization controller.

II. Periodic Random Polarization Scrambling

Traditional polarization scrambling schemes always use different driving functions on each section of polarization controllers (either electrooptic or fiber-squeezer based) and can move the output SOP over the Poincarè sphere, while they may suffer from uncertainty (randomly nonrepeatable) from measurement to measurement, and it is difficult to uniformly distribute the output SOP over the sphere. To obtain the most efficient PMD monitoring and compensation, it is desired that the output (scrambled) SOPs are uniformly distributed and

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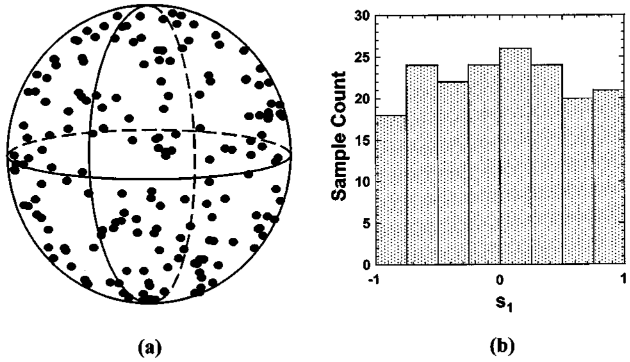


Fig. 1. Periodic random polarization scrambling. (a) Measured output SOPs of $\sim$200 samples on the Poincarè sphere. (b) Distribution of one of the Stokes parameters (s_1).

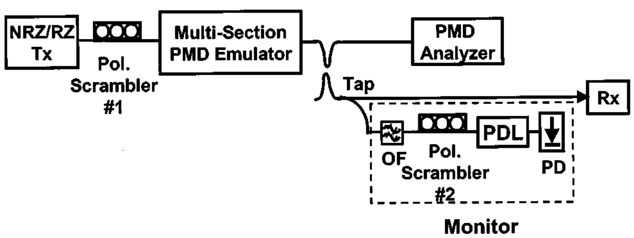


Fig. 2. Diagram of the proposed DGD monitor; OF: Optical filter; PD: Photodiode.

repeatable. In order to realize uniformly distributed SOPs for any arbitrary unknown input SOP, we analyze the polarization transfer matrix and derive a random polarization transformation matrix as [11]

$$\mathbf{P} = \begin{bmatrix} \sqrt{\gamma}\exp(i\theta) & -\sqrt{1-\gamma}\exp(-i\phi) \\ \sqrt{1-\gamma}\exp(i\phi) & \sqrt{\gamma}\exp(-i\theta) \end{bmatrix}$$

where $0 \le \gamma \le 1$, $-\pi \le \theta \le \pi$, and $-\pi \le \phi \le \pi$ are statistically independent and uniformly distributed random variables.

Since we use fiber squeezer-based polarization scrambler for our experiments, by multiplying the matrices of the three-section squeezer [12] and comparing the product with the matrix $\mathbf{P}$, we find that in order to obtain uniformly distributed SOPs, we can distribute θ_1 and θ_3 uniformly between $[-\pi, +\pi]$, while distributing $\sin\theta_2$ uniformly between $[-1, +1]$. (θ_1, θ_2, and θ_3 are the corresponding phase retardation values of the three sections.) A series of random control voltages are generated according to this distribution and we periodically apply them to the polarization scrambler to perform random polarization scrambling. Fig. 1(a) plots the output SOPs on the Poincarè sphere ($\sim$200 samples) for an arbitrary input polarization state, while Fig. 1(b) shows the uniformity of one of the Stokes parameters, in this case s_1.

III. DGD Monitor Configuration

Fig. 2 shows the DGD monitoring experimental setup. A tunable laser is externally modulated with two cascaded electrooptic (EO) modulators to achieve nonreturn-to-zero (NRZ) as well as half duty-cycle RZ intensity modulation at

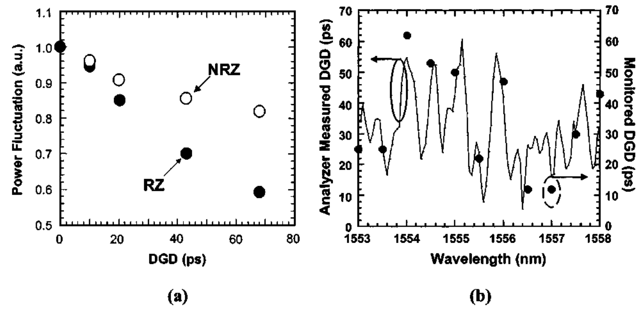


Fig. 3. DGD monitoring results. (a) Monitored DGD versus normalized power fluctuation for 10-Gb/s NRZ and RZ data formats using a variable DGD element. (b) Comparison between the DGD values of the 12-section PMD emulator given by our monitor (solid circles) and a commercial PMD analyzer (dashed line).

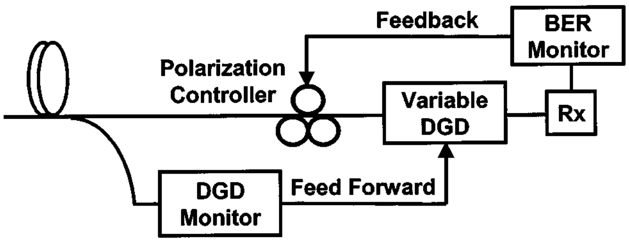
10-Gb/s ($2^{23}-1$ PRBS). A 50-kHz polarization scrambler that can produce periodic polarization scrambling over a series of uncorrelated SOPs as discussed in Section II is placed after the transmitter. The PMD of the link is emulated using a PMD emulator that consists of 12 sections of polarization-maintaining (PM) fiber to emulate $\sim$33-ps average PMD by randomly rotating the relative orientation of each PM fiber section [13].

The concept of DGD monitoring at the receiver is described as follows. By tapping off some power of the signal at the receiver and filtering out the desired data wavelength, we monitor the instantaneous DGD using a polarization scrambler followed by a fixed 0.8-dB PDL element. The receiver polarization scrambler is used to time-average the polarization coupling between the link PMD and the PDL element. If the PMD value along the link increases, then the instantaneous DOP will decrease, thus causing a decrease in the measured RMS power fluctuation value through the PDL element. This power fluctuation can be determined within one polarization-scrambling period, which could be on a submillisecond time scale. In this monitor configuration, locking or synchronizing of the transmitter and receiver polarization scramblers is not necessary, and a simple polarizer can be used instead of the PDL element. In addition, without the first polarization scrambler at the transmitter, the DGD monitoring module works as a traditional effective PMD monitor or alternatively, as a DOP monitor.

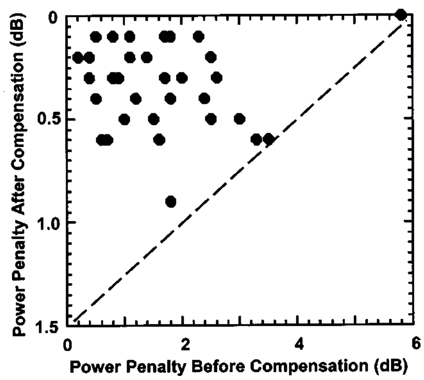
IV. DGD Monitor Results

For a simple variable DGD element, Fig. 3(a) shows the measured power fluctuation variation after normalization as a function of instantaneous DGD for both NRZ and RZ 10-Gb/s data at 1555 nm. As the DGD value changes from 0 to 70 ps, the monitor signal dynamic range due to PMD is $\sim$20% for NRZ and $\sim$40% for RZ. For NRZ format, consecutive "1" bits are not degraded by PMD, hence the NRZ monitored power fluctuation is less sensitive to PMD changes than is RZ.

Furthermore, in the case of all-order PMD involved, we use the 12-section PMD emulator and compare in Fig. 3 (b) the normalized instantaneous DGD of our monitor (RZ data stream) with the measured DGD using a commercial PMD analyzer. Our monitored DGD values agree well with the measured values. The difference between monitored and measured values for some points is likely due to higher order PMD effects.



(a)



(b)

Fig. 4. (a) Illustration of first-order PMD compensation at the receiver for a 10-Gb/s NRZ channel using monitored DGD as feedforward information. (b) Compensation results.

These results show the utility of this technique for a wavelength-division-multiplexing (WDM) system, in which an optical filter can be scanned across an entire wavelength range and the instantaneous DGD values can be rapidly acquired, thus PMD compensation can be accomplished using one DGD monitor and N traditional PMD monitors for an N-channel WDM system.

V. Feed-Forward PMD Compensation

In order to show the effectiveness of this DGD monitoring scheme, we further demonstrate first-order PMD compensation for a 10-Gb/s NRZ data channel. As shown in Fig. 4(a), the PMD compensator is composed of a polarization controller followed by a variable DGD element up to 70 ps. Since polarization scrambling is applied at the transmitter to facilitate DGD monitoring and PMD compensation, a variable DGD compensator must be applied here to exactly cancel the monitored DGD along the link [9]. The monitored DGD is used as a feedforward control signal to set the DGD value of the variable compensator, and the BER information at the receiver is used as the feedback signal to manually adjust the polarization controller. Although this configuration, i.e., combining feedforward and feedback control, may seem a little complicated, it can actually reduce the complexity of the control algorithm, which is one of the major challenges for PMD compensation. The PMD of the link is emulated using the multisection emulator. Compensation results down to a $\mathrm{BER} = 10^{-9}$ are shown in Fig. 4(b). The residual penalties after first-order compensation are likely due to either the higher order PMD-induced monitoring mismatch or higher order PMD itself.

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